

Exploring Stanislavskian Performance for Agent-based Nonplayer Characters through Defeasible Logic

Kyle Mitchell

Joshua McCoy

kdmitch@ucdavis.edu

jamccoy@ucdavis.edu

University of California, Davis

Davis, CA, USA

ABSTRACT

Common approaches to behavioral artificial intelligence, like behavior trees, utility-based approaches, and machine learning are often lacking with regards to expressivity of the decision-making process. Behavior trees are too rigid; utility-based approaches are focused on outcomes, not process; and machine learned agents seek only to maximize reward and minimize loss. A combination of AI approaches, however, may provide more nuanced, visible, and coherent processes that can be realized through the performance of character behaviors. In this extended abstract, we present Viv: a system that blends defeasible logic programming with dynamic behavior trees. To our knowledge, no extant works have bridged the gap between defeasible reasoning and action, particularly in regards to the performances of virtual nonplayer characters. We begin by explaining the design of our system, which was heavily influenced by the Stanislavski method of acting. We then describe the technical implementation of our system. Finally, we describe future directions of research that could provide benefit to, or benefit from, our system.

CCS CONCEPTS

• **Applied computing** → **Performing arts**; **Computer games**;
• **Computing methodologies** → **Nonmonotonic, default reasoning and belief revision**; **Logic programming and answer set programming**.

KEYWORDS

computational performance, defeasible logic, logic programming, nonplayer characters, video games

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1 INTRODUCTION

There are myriad ways to build behavioral intelligence into virtual agents, ranging from finite state machines to custom scripting languages. A great amount of work has been done with regards to behavior trees, utility-based approaches, and machine learning methods involving reinforcement [8] [22] [3]. These common implementations of behavioral intelligence, however, each taken on their own, lack complex expressivity and nuance with regards to the decision-making process of agents. For instance, behavior trees, while transparent, struggle with exceptions; and agents driven by utility can make confusing choices due to a lack of clear reasoning. Similarly, an agent trained through reinforcement learning simply maximizes reward, and cannot say why it chose one action over another, especially in situations where all possible actions lead to a similar outcome [20]. In sum, it is difficult to maintain flexibility of character behavior, the coherency of those behaviors, and the explainability of the performance as a whole in equal parts.

To address this, we present “victoria in victus,” or Viv for short: a system that blends dynamic behavior trees with defeasible logic, inspired by the Stanislavski method of acting. Defeasible logic, based in argumentation theory [17] [5], helps us to better question character behavior, and therefore represent character performance. Computational models of the process have been built: in classical logic-based approaches [4] [2] and in defeasible logic programming [6]. Defeasible logic provides the capability of asking questions, answering those questions, and providing arguments for those answers: a process that is well-complemented by Stanislavski’s method. Some of Stanislavski’s teachings have been applied to other areas of computer science, like in coding for human animation [11], and they have also seen recent application toward the direct control of characters’ performance with regards to their emotions, as with the *Puppitor* system [10].

To the best of our knowledge, no work or system exists that directly connects defeasible logic programming to the actions or behaviors of virtual characters. Thus, the primary concern of this work is the exploration of defeasible logic, combined with behavior trees, as a central component of flexible, coherent, and explainable character performance. We begin by describing the design of our system with respect its Stanislavskian inspiration. We then offer a more technical description. Finally, we comment on several promising research directions in which this work could be taken.

2 SYSTEM DESIGN

What about the process of contextual defeasibility is most important, and how do the teachings of Stanislavski help us model this process?

In this section, we offer answers to those questions by visiting three critical components of Stanislavski’s method that we wished to see replicated in our system: supertasks, tasks, and the “Magic If.”

Our system is meant to enable characters to perform in a way that reflects a prominent framework for human actors: Stanislavski’s method. At its heart, the method is centered around the concept of “experiencing the role.” Stanislavski wanted his actors to experience feelings and emotions analogous to those that the character experiences. Stanislavski posited that only when the character and the actor act in complete alignment, can they pursue the “supertask,” or the task that operates at the scale of the drama as the whole. A supertask does not simply reflect a character’s personal desire. Rather, it is a more unifying objective that ties together all of their actions throughout the play. It represents the character’s journey, from beginning to end. Thus, the supertask is the entry point into our system. From the supertask, tasks are created.

Stanislavski teaches that a performance is punctuated by the completion of a sequence of tasks. A task is a problem to overcome, embedded in the given circumstances of the performance. Tasks can be equated to “obstacles,” as defined by more modern developments of Stanislavski’s system [7]. Actors often create these tasks by asking questions like: *What does my character want? What do I do in order to get it? If things change, what do I do then?* Critically, tasks are the results of questions concerned with context. In our system, tasks are referred to as *behaviors*.

One of the most resonant pieces of Stanislavski’s system is his cardinal exhortation to employ the use of the “Magic If.” This phrase refers to the ability to imagine oneself in a set of “given circumstances,” and respond to them in the form of actions that the character would take. We argue that this process is akin to that of reasoning by contextual defeasibility: the “Magic If” reflecting a query being given to a knowledgebase; the given circumstances taking the place of stateful information of a virtual world at the moment of the query; and the changing of the circumstances to the changing of states within that world, perhaps brought on by the agent’s own behavior.

3 TECHNICAL DESCRIPTION

There are four main components of our system, shown in Fig. 1: a defeasible logic knowledgebase; an agent within the ABL reactive planner; the game world within which we tested our system; and the central management system, named Viv. In this section, we focus on the defeasible logic knowledgebase and Viv in particular. For more thorough descriptions of the ABL agent, we refer to work done by Mateas and Stern [12] [13]; for more on the agent’s incorporation into the game world, we refer the reader to previous work done by Mitchell and McCoy [16].

3.1 Defeasible Logic Knowledgebase

A defeasible logic program consists of a set of facts, strict rules, and defeasible rules. Examples of syntactic use are shown below.

- **Facts:** *Penguin(Opus)*.
- **Strict Rules:** *Bird(X) ← Penguin(X)*.
- **Defeasible Rule:** *Bird(X) -< CanFly(X)*.
- **Query:** *Bird(Opus)*

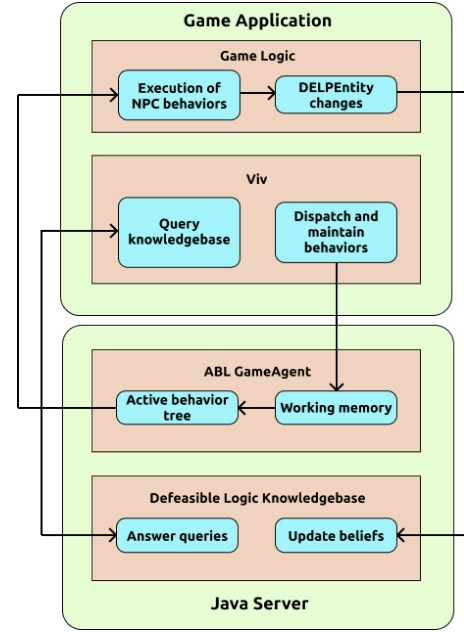


Figure 1: An architectural overview of our system. There are two main divisions: components contained within the game application (the game logic itself and Viv) and components contained within the Java server (the ABL GameAgent and the defeasible logic knowledgebase).

A query is expressed in the same way as a fact, and the result of a query may be one of four values: *YES*, if the given literal is warranted; *NO*, if the complement of the literal is warranted; *UNDECIDED*, if neither the literal nor its complement is warranted; and *UNKNOWN*, if the literal does not belong to the signature of the knowledgebase.

With regards to our implementation, we adopted an approach similar to that described in previous work done by Agis et al. [1]. We first wrote a Java-based API exposing functionality of a larger logic programming library known as the Tweety Project [21]. This API, encapsulated in a class called the *DelpHandler*, allowed us to create, update, and otherwise manipulate a defeasible logic knowledgebase that can operate within the context of a local Java server.

3.2 Viv

At the heart of our system is Viv, short for the Latin phrase “victoria in victus,” or, roughly translated, “victory in defeat.” It is responsible for querying the defeasible logic knowledgebases of characters; dispatching the appropriate behaviors for characters to take based on a given supertask; evaluating the assumptions on which those behaviors are based; and ultimately, choosing the behavior that best fits the given circumstances.

3.2.1 Data Structures. A *Supertask* contains both a set of associated *Behaviors* and an array of internal evaluations of those *Behaviors*. A *Behavior* represents an action plan that could reasonably help the character achieve the *Supertask* with which it is associated. Every *Behavior* is associated with a set of underlying *Assumptions*: logical

statements presumed to be fact that can be transformed into a query. Critically, an *Assumption* relies upon the *DELPEntity* component of the character making the *Assumption*. A *DELPEntity* component represents information that characters should reasonably know and can infer throughout the game, in the form of facts, strict rules, and defeasible rules.

3.2.2 Phases of Operation. A *Supertask* is first given by an auxiliary system (currently, through game domain-specific logic). Upon receiving a *Supertask*, Viv compiles a list of possible *Behaviors* that can be used to achieve it. These *Behaviors*, in turn, spawn *Assumptions* that must be validated.

Viv then performs an initial evaluation. Every *Assumption* belonging to each *Behavior* is validated; that is, it is transformed into a query that can be interpreted by the defeasible logic knowledgebase. If the answer returned by the knowledgebase is *YES*, the assumption is valid; if the answer is *NO*, the assumption is invalid; if the answer is either *UNDECIDED* or *UNKNOWN*, the assumption is marked as uncertain. Each *Behavior* stores the most current evaluation of its underlying *Assumptions* in an array.

Viv then processes the *Supertask* and selects particular *Behaviors* for dispatch to the ABL GameAgent. Its process of selection occurs according to the following algorithm:

- (1) *Behaviors* built on one or more invalid assumptions are removed from consideration.
- (2) *Behaviors* built on only valid assumptions are selected. If there are none, continue on to the following step.
- (3) *Behaviors* built on any number of uncertain assumptions are selected.

The selected *Behaviors* are then dispatched to the ABL GameAgent. All *Behaviors* built on only valid assumptions will execute to completion, but any *Behaviors* built on uncertain assumptions will be paused. Any *Behaviors* currently paused can be re-evaluated in response to a change to the acting character's *DELPEntity* component. Viv may dispatch messages to the ABL GameAgent such that it either resumes paused *Behaviors* due to a change in its associated *Assumptions* becoming valid, or discontinues it due to its associated *Assumptions* becoming invalid.

4 FUTURE WORK

Given the system in its current state, we have identified a few thrusts of future work: linking Viv to a system that can dynamically define supertasks; deepening the influences of Stanislavski's work in acting with regards to character motivation; and extending the capabilities of the system to include more specific, higher-fidelity control of characters' bodies.

4.1 Formation of Supertasks

At the moment, Viv can only process a supertask that is given to it. Thus, in our testing, the supertask constructed was a static object that was fed directly to Viv. We chose to do this for two main reasons: (1) Classically, the goal itself is often separated from the concrete attainment of said goal; and (2) We did not want to overcomplicate the system so early in its development. We do, however, see a potential relationship between systems that can form and set these high-level goals based on character design (i.e.,

personalities, traits, and the like). One such candidate system to incorporate into this work is *Comme il Faut* [14] [15].

4.2 Stanislavskian Motivations

Stanislavski's teachings might be able to help better model an agent's core motivations. Basic survival is often cited in games as a core motivation for NPCs, but more complex motivations can be layered in. For instance, a security guard agent might prioritize protecting civilians (programmed within the ABL GameAgent) but also have a secondary motivation to uphold the law (encoded in defeasible logic rules).

With better defined motivations, we might also consider how those motivations can be translated into action through defeasible logic. This could be realized in the form of shifting priorities based on context, e.g. the weight of a defeasible logic rule (protecting civilians) could be overridden by another (flee imminent danger) if the in-game situation escalates.

4.3 Fidelity of Embodiment

To express the inner state of a character through the body, in a nonverbal way, is one of the most critical goals of an actor. We believe that, ultimately, nonplayer characters in virtual worlds should share this goal. There has been some recent work by this community toward such efforts [18]. With appreciably considerable effort, Viv could be extended to control such bodily movement, given more suitable, higher-fidelity character models, and an underlying model of inner character state, encompassing not only knowledge, but emotion. We recognize the large body of research on modelling emotion for use in virtual agents, and hope to incorporate the findings from recent works in this community [19] [9].

5 CONCLUSION

In this work, we have presented the beginnings of Viv: an AI system that blends defeasible logic programming with dynamic behavior trees, and heavily influenced by the works of Constantin Stanislavski. Our system represents a step towards more holistically flexible, coherent, and explainable character performance. We first described our system design with regards to its Stanislavskian influence. We then provided a technical description of our system, focusing on the defeasible logic knowledgebase and Viv itself. Finally, we commented on some directions for future work, including more native formation of supertasks; the programmatic representation of complex motivations; and fidelity of embodiment with characters controlled by Viv.

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